
Internship Report

Antoine LI

Mines Paris | PSL / MVA ENS Paris-Saclay

Stage : Diffusion-based forward model for articulatory
trajectories generation

COML - ENS Ulm

April 2025 - September 2025

October 17, 2025

Contents

1	Introduction	3
1.1	General context of the internship	3
1.2	Research problem addressed	3
2	Preliminary work	5
2.1	Literature Review	5
2.2	SPARC	6
2.3	Grad-TTS	8
2.4	About Text-to-Speech	10
2.5	Research Directions	11
3	SPARC multilingual AAI probing	11
3.1	Electromagnetic Articulography (EMA) datasets	12
3.2	Methodology	13
3.3	Results and Discussion	13
4	Monospeaker articulatory-based TTS	14
4.1	Data	14
4.2	Methodology	15
4.3	Results and Discussion	18
4.3.1	Articulatory Performance Analysis	18
4.3.2	Speech Synthesis Quality Analysis	19
5	Multispeaker Multilingual Articulatory-based TTS	21
5.1	Data	21
5.2	Methodology	22
5.3	Results and Discussion	23
5.3.1	Articulatory Performance Analysis	24
5.3.2	Speech Synthesis Quality Results	24
5.4	Discussion	25
5.5	Further discussion	27
6	Conclusion	28

Brief Summary

Human voice generation is based on the movement of various organs in the vocal tract. The spatial displacement of these points of interest constitutes an articulatory trajectory. Their interest lies in their phonological proximity, interpretability, and universality. However, obtaining such data coupled with audio was difficult until recent work [7] demonstrated that it is possible to approximate them from audio using large SSL models such as wav2vec. In addition, articulatory encoding is relevant enough to obtain high-quality audio reconstruction using a HiFi-GAN. This opens up the possibility of generating controllable articulatory trajectories from text or rather phonemes, the latter having the advantage of not being directly dependent on language. To our knowledge, there is no such TTS model using articulatory space as an intermediate space. Our idea was to adapt the Grad-TTS diffusion-based model [18] by using phonetic features as input and replacing the spectral space (mel-spectrogram) with the articulatory space.

1 Introduction

1.1 General context of the internship

I had the pleasure to be part of the CoML (Cognitive Machine Learning) team, based in the Department of Cognitive Studies at the École Normale Supérieure (ENS) in Paris. The team is a joint group involving EHESS, ENS, CNRS, and Inria, and brings together researchers at the crossroads of cognitive science and computer science under the supervision of Emmanuel Dupoux.

While recent advances in machine learning have achieved impressive results, there are still many areas where even very young children outperform artificial systems such as learning language without supervision, reasoning logically, or adapting flexibly to new situations. CoML's goal is to better understand these human abilities and to design algorithms that reproduce them. The research combines theoretical aspects (developing efficient and scalable algorithms) with empirical validation, by comparing model predictions with behavioral or neuroscientific data. This dual approach makes the team's work relevant both for artificial intelligence and for cognitive science.

In particular, the team has a strong focus on understanding speech and language acquisition and is interested in developing human-based models.

1.2 Research problem addressed

Current speech models, particularly self-supervised learning (SSL) models, operate largely as black boxes that have grown increasingly disconnected from the underlying human mechanisms of speech production. While these models achieve impressive performance on downstream tasks through data-driven deep learning strategies, they lack interpretability and provide little insight into why they work and how they can be improved. Current speech synthesis models suffer from similar limitations. Indeed, these models typically operate directly in the acoustic domain and consequently learn more about the data they are trained on than the speech production process. This leads to limited controllability over their output with poor ability to generalize over varying

speaking modalities such as language, speaker identity, prosody, pronunciation, etc. In this regard, [articulatory trajectories](#) represent a relevant field of research.

Articulatory trajectories refer to the physical movements of speech articulators such as the tongue, lips, and jaw during speech production. Unlike acoustic features that represent the result of speech production, articulatory trajectories are its very source and represent the underlying physical speech intent. Using articulatory features to model speech thus offers multiple advantages. Indeed, they are more interpretable, controllable, and their universal principle suggests a potential to generalize better across speakers and languages. However, articulatory trajectories need to be acquired with electromagnetic articulography (EMA) methods, which suffer from a severe lack of data, explaining the current preference for acoustic features.

Recent advances in the field [8] have demonstrated that latent representations learned by self-supervised learning (SSL) speech models encode not only acoustic properties but also articulatory information. These representations show strong correlations with articulatory trajectories and were used to build a powerful articulatory-based coding system of speech named [SPARC](#) (Speech Articulatory Coding) [7] showing interesting results. First, the possibility to generate large-scale data using an Acoustic-to-Articulatory inversion (AAI) system built with a simple linear transformation on top of the SSL features. In addition, the generated articulatory trajectories are relevant enough to reconstruct a high-quality speech signal using a HiFi-GAN synthesizer [13]. Moreover, it has been shown that latent representations of a single speaker can generalize to unseen speakers and languages [7][9], enabling the possibility of controllable speech synthesis with an articulatory bottleneck.

On the other hand, deep generative models, such as diffusion probabilistic models (DPMs), have lately gained traction in many machine learning tasks, across text, image, and speech modalities. In particular, DPMs have been used to produce high-quality audio by reconstructing the acoustic features in Grad-TTS [18] or the raw waveform in DiffWave [14]. Given their generative nature, DPMs have the potential to produce controllable trajectories in articulatory space based on discrete representations.

The research problem addressed in this internship is the development of a diffusion-based forward model for articulatory trajectories. Specifically, the goal is to generate controllable, latent articulatory kinematic trajectories from discrete phonological configurations, such as [IPA](#)-like, ternary-valued phonological features. While character or phoneme inputs are generally used in speech synthesis, we prefer using phonological traits for better universality and generalization capacities over rare languages and unseen phonemes. Successfully solving this problem would contribute to articulatory-informed speech synthesis and open avenues for cross-lingual and cross-speaker controllable Text-to-Speech (TTS) systems.

2 Preliminary work

2.1 Literature Review

The main challenges refraining current research from exploiting the potential of articulatory trajectories are the lack of data and its heterogeneity. Indeed, the physical acquisition procedure using Electromagnetic Articulography [EMA](#) method is costly and complicated. Moreover, these datasets are of diverse quality with varying recording setups and varying vocal tract geometries across speakers. Regarding those issues, Cho et al. achieved remarkable work. Not only have they shown evidence that SSL speech models representations can be used to build efficient Acoustic-to-Articulatory Inversion ([AAI](#)) systems [8], they also proved high cross-lingual and cross-speaker transferability [9]. Given that, it was possible to fine-tune an SSL-based [AAI](#) inversion on a single speaker articulatory space with little EMA data and without loss of generality. Indeed they used MNGU0 [EMA](#) dataset [21] as a universal articulatory template space upon which they built a Speech Articulatory Coding ([SPARC](#)) framework [7]. SPARC encodes any speaker’s speech into MNGU0-speaker space and reconstructs it successfully, showing evidence that the template articulatory space has the ability to meaningfully encode speech content independently of the speaker. We thus have both a proven, goal articulatory space for our phoneme-to-articulatory synthesizer and a process to generate articulatory targets at will from speech samples thanks to SPARC SSL-linear based encoder.

With most research focusing on acoustic-to-articulatory inversion or articulatory-to-acoustic synthesis, limited direct text-to-articulatory work exists. However, the text-to-speech field has been particularly active over the recent years thanks to the emergence of neural architectures, with the vast majority of models having a first *feature generator* part, converting text to time-frequency acoustic features (such as mel-spectrograms). This spectral intermediate representation has proven highly effective, enabling models like Tacotron2 [25], FastSpeech [20], or Glow-TTS [11] to be increasingly fast and realistic. Lately, diffusion-based approaches began to gain traction in the speech synthesis field. Indeed, given their probabilistic nature Diffusion Probabilistic Models (DPMs) [26] are particularly suited for the one-to-many mapping generation task of TTS models. Moreover, their evolution has led to computationally efficient methods [27] that were notably applied for TTS time-frequency feature generation in the Grad-TTS paper [18]. This gives us confidence about the ability to adapt Grad-TTS method for articulatory features generation, especially since the articulatory space is lower dimensional.

While the text-to-articulatory generation is the main focus of the internship, we also have good hopes to extend it to a TTS model. In order to do so, we need an efficient way to perform articulatory-to-speech synthesis (more commonly referred to as [articulatory synthesis](#)). Despite some early stage works, the current articulatory-to-speech methods mostly inherit from the traditional TTS vocoders. These vocoders usually generate audio waveforms from spectral features through neural architectures like HiFi-GAN [13] or WaveGlow [19]. The first articulatory methods tried to map articulatory features to acoustic features, but a recent study [29] showed that high-intelligibility could be obtained directly from articulatory features using a HiFi-GAN architecture. The fact that both Grad-TTS and SPARC use a HiFi-GAN as their vocoder comforts us in choosing it for speech

synthesis. Moreover, the latter already features a speaker conditional HiFi-GAN to compose with the fact that our articulatory space is speaker-agnostic.

The previous paragraphs present the main supports and motivations to our work. However, one of the main difficulties in speech synthesis is to evaluate the results in an objective and efficient way. For the articulatory trajectories, the Pearson Correlation Coefficient (PCC) is widely used for its ability to compare relative variations. However, having an objective and reliable metric to assess speech quality has been a long-standing problem. The Mean Opinion Score (MOS) is a subjective measure of speech quality, obtained by asking human listeners to rate the perceived naturalness or intelligibility of samples on a numerical 1–5 scale. It is inherently costly to obtain and far from being an absolute reliable score [15]. But despite its limitations, it remains the standard evaluation metric. To tackle the cumbersome MOS collection, a machine-evaluation trying to replicate the human MOS score has been developed as UTMOS [23], with limited reliability. For interpretability and comparability purposes, we sometimes choose to use aligned phonemes rather than a raw text input, thus bypassing the burden of phoneme duration prediction and eliminating a factor of uncertainty.

Given how fundamental SPARC and Grad-TTS are to our work, the following subsections are dedicated to describe their contribution as well as technical specifications in greater detail.

2.2 SPARC

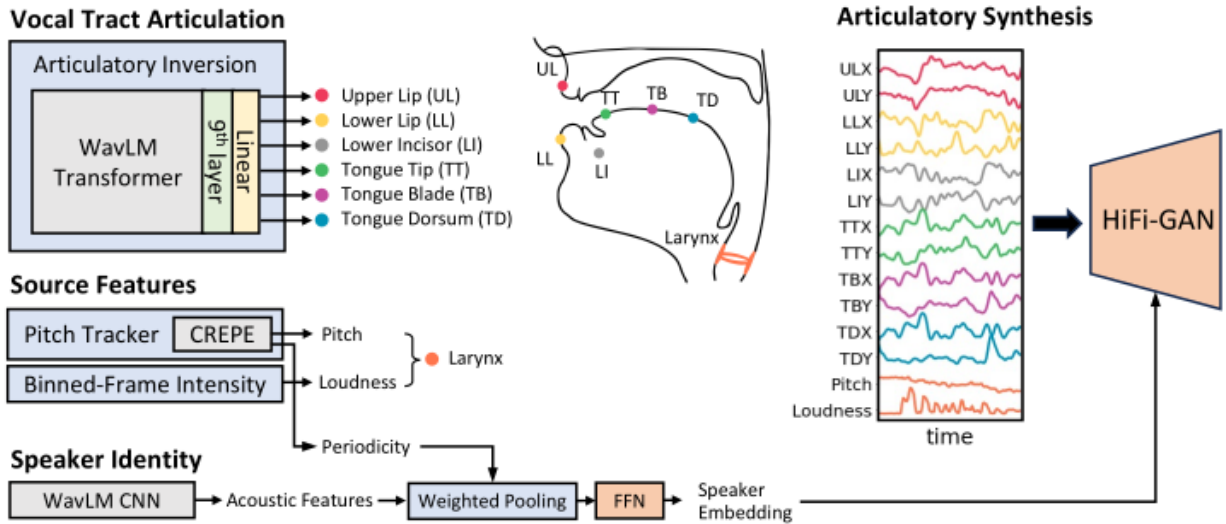


Figure 1: From SPARC paper. Pipeline of articulatory analysis and synthesis. The articulatory analysis is composed of vocal tract articulation, source features, and speaker embedding, which are then fed to the synthesizer (HiFi-GAN) in the synthesis pipeline. The modules colored with orange (FFN and HiFi-GAN) are updated while training the synthesis model and other modules are fixed.

SPARC Articulatory Space. In the SPARC framework, the articulatory space is actually made of two components: vocal tract articulation and source features which, together, represent the speaker-agnostic speech content. Importantly, the approach goes beyond relying solely on (EMA) data, that, although informative, is not totally sufficient. In this wide-sense definition of "**articulatory features**", EMA articulator coordinates are combined with pitch and loudness signals totalling up

to 14 channels (12 EMA + 2 source) at a 50 Hz sampling rate (to match SSL rate). Moreover, an additional speaker embedding module is necessary to retrieve speaker-specific voice texture for speech resynthesis.

The three speech coding blocks are:

- **Vocal Tract Articulation:** Captures purely phonological content. The obtained articulatory trajectories are supposed to be speaker-agnostic; in other words, they would be the trajectories extracted if the exact same speech was pronounced through MNGU0-speaker's mouth. This ensures phonetic information is preserved independently of anatomical variations, making it possible to generalize across multiple speakers. This AAI inversion is learnt separately by training a single linear layer from WavLM [6] 9th-layer features on a single-speaker EMA dataset (MNGU0).
- **Source Features:** Captures glottal excitation and prosodic variation via the extraction of pitch and loudness from the source audio. It is done using CREPE [12] to infer pitch from speech, while loudness is measured by the average of absolute magnitudes of waves for every 20 ms bin.
- **Speaker Identity:** Intended to extract the missing structural and anatomical characteristics (e.g., vocal tract length) that shape an individual's unique voice quality. A speaker encoder, trained jointly with the synthesizer, reintroduces this information by learning a Feed-Forward Network (FFN) on top of acoustic and periodicity features. These features are respectively extracted using frozen WavLM CNN part and CREPE. This is motivated by previous studies showing that speaker-specific information is located in the CNN outputs of SSL models. The periodicity features are here used to grant more attention on periodic portions of signal, in the weighted pooling process of acoustic features that outputs a fixed-size 1024-dimensional speaker vector. The FFN is then used to learn a task-specific 64-dimensional speaker embedding.

SPARC articulatory space configuration is intended to be universal and completely disentangled from speaker-specific characteristics while capturing purely phonological pronunciation content. Indeed, the main hypotheses are that vocal tract structures can be aligned via an affine transformation and that the SSL-extracted features are speaker-agnostic. While it holds for EMA articulators, one can argue that source features actually include non-phonological content that will be hard to predict in a TTS task (pitch, loudness). One can also argue that the reconstruction might not be purely articulatory given that the speaker embedding is actually built from the source speech signal itself.

SPARC Articulatory Synthesis. For the articulatory synthesis step, an adapted HiFi-GAN is used as a vocoder, trained to reconstruct speech waveforms at a 16 kHz sampling rate directly from articulatory features. To ensure speaker-specific rendering, the model is conditioned on the speaker embedding using Feature-wise Linear Modulation (FiLM [17]), which adjusts the convolutional layers according to speaker characteristics. Training follows the standard HiFi-GAN setup, combining a "mel-spectrogram" (here articulatory trajectory from audio waveform AAI)

reconstruction loss with multi-period and multi-scale discriminator losses, thereby enabling the generation of high-quality, natural-sounding speech.

Training. The training dataset used was LibriTTS-R, a high-quality English dataset of audiobooks reading (555 hours of training). It is important to note that only the HiFi-GAN and the speaker embedding FFN are learnt. The articulatory coding being pre-learned and fine-tuned by the SSL-linear block. We can assume that they used the same training method as the HiFi-GAN featuring an adversarial loss, a feature-matching loss, and a mel-spectrogram loss. It is important to note that it requires the prediction and the target being of the same length. The main SPARC version provided by the authors is the English one learnt on LibriTTS-R; however, they also provide a multilingual HiFi-GAN checkpoint allegedly learnt on a multilingual dataset. Unfortunately, it seems to be less stable and of lesser quality when using it.

Intended Use. The main use of SPARC will be to use its AAI encoding to create custom datasets with coupled audio and SPARC-approximated articulatory features. Besides, its HiFi-GAN can be used as a free vocoder to perform articulatory-to-speech synthesis for our TTS model.

2.3 Grad-TTS

Grad-TTS originated from the observation that people already tried to use diffusion in the vocoder part but not in the feature generation part of TTS models. The authors employed score-based diffusion models [27] to iteratively refine Gaussian noise into high-quality mel-spectrograms through a reverse diffusion process. This method has the advantage of producing diverse, natural-sounding speech with an adjustable inference speed. Its general principle is detailed in Fig. 2.

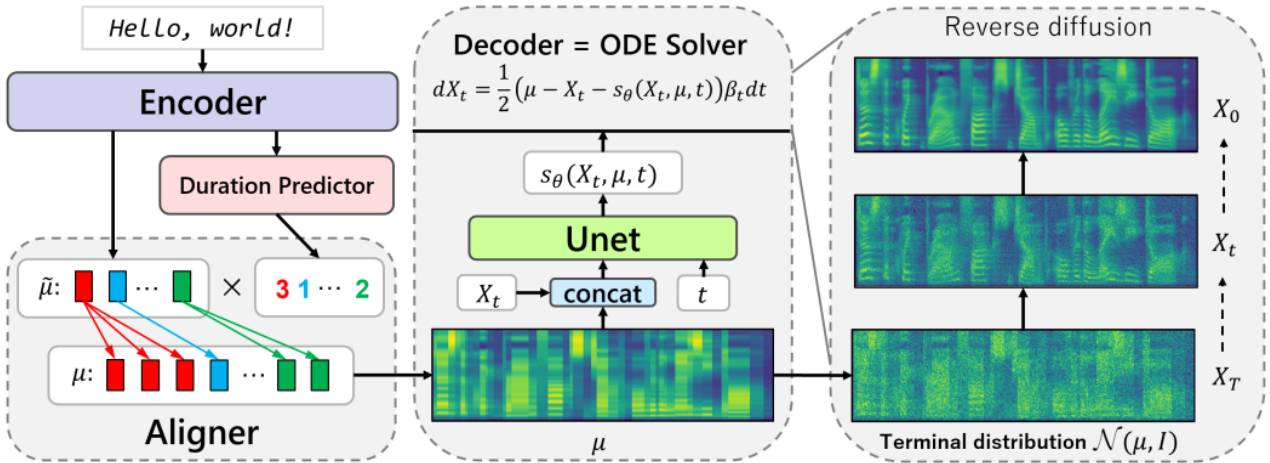


Figure 2: From Grad-TTS paper. Grad-TTS inference scheme

Encoder The goal of the encoder is to produce a gross approximation of the expected output from a text or phoneme input. Since the main challenge of TTS consists in converting discrete inputs into continuous features, the encoder is here to provide a starting point for the decoder. The architecture borrowed from Glow-TTS [11] is a Transformer-based one that produces a mel-spectrogram for each token of the input sequence. Their durations are then adjusted by increasing their multiplicity using a duration predictor. We will not detail its functioning as we actually do not

want to use it, because we have no grip on its output length and it heavily relies on the training data (speaker, language, content type...). We would rather ignore it when having aligned inputs available.

Decoder The decoder was designed in a way to efficiently produce realistic mel-spectrograms. The main innovation brought by the Grad-TTS authors relies on the possibility to condition generation by the coarse-grained encoder output (see Fig. 3). This was done using an adaptation of score-based diffusion methods [27] that centers the noising process on the encoded prior instead of zero. The score estimator employed is an adapted U-Net architecture [22] [10] processing the 2D spectrograms as images.

The waveform is then reconstructed from the decoded mel-spectrogram using a HiFi-GAN, but we are not really interested in it.

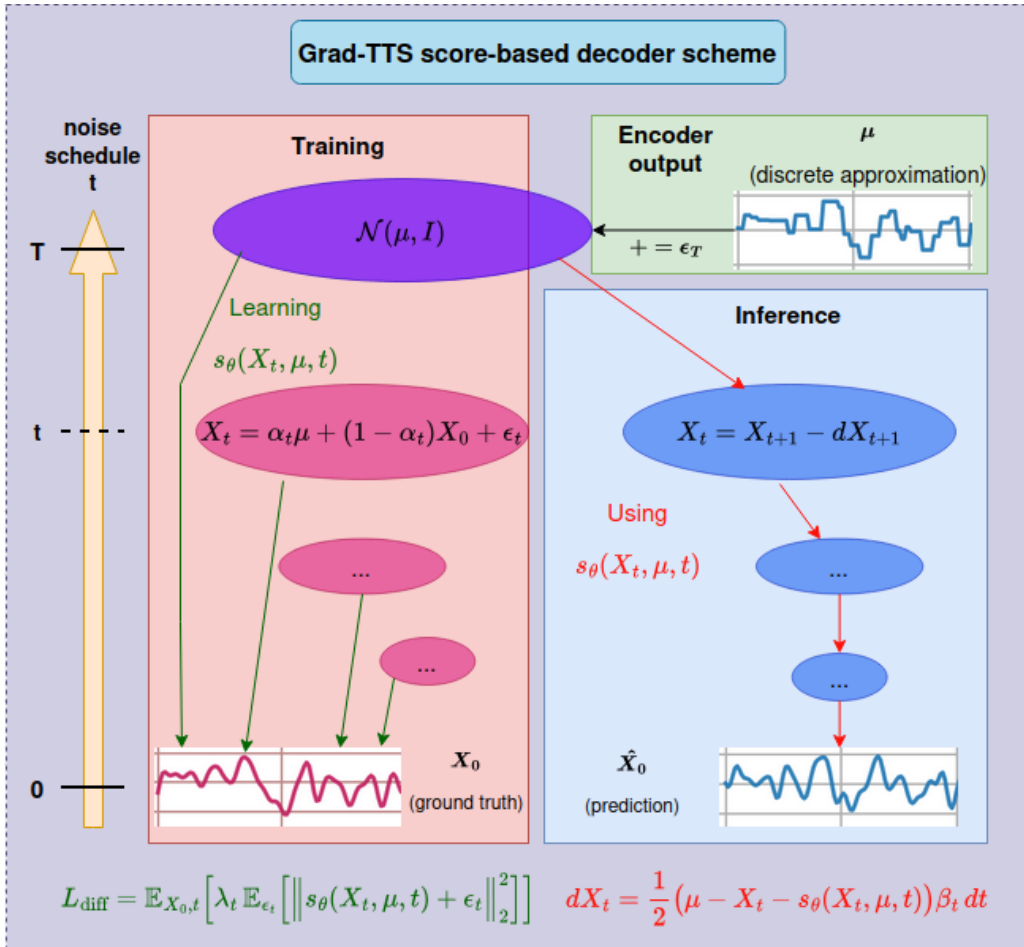


Figure 3: Grad-TTS score-based diffusion process guided by a discrete prior approximation

Training LJSpeech dataset containing around 24 hours of high-quality, English female voice recordings was used to train Grad-TTS. The model optimizes on three losses: a prior loss (comparing phoneme-level mel-spectrogram features), a duration loss, and a diffusion loss. The training spanned over 2300 epochs for 1.7m iterations.

Intended use Using the fact that articulatory features are two-dimensional (considering time), the model can easily be adapted by replacing the spectral channels with articulatory ones. However, we have to keep in mind that they probably have less spatial structure than images, but their lower dimensionality (14 compared to 80) and their phonetic grounding suggest easier learning. About input handling, Grad-TTS opted for a learned embedding setup on a hybrid character/phoneme

vocabulary. While it answers industrial needs better thanks to its ability to process any Latin alphabet text, it restricts its soundness. It is even more annoying since their text-to-phoneme conversion, based on a dictionary, has a poor conversion rate, leading to a high amount of letter residuals within the input. In order to build a more general model, we will instead focus on fully phonemized input and use phonetic features instead of a learned embedding that has the downside of being highly data-dependent. Lastly, despite its main version being mono-speaker, Grad-TTS also offers a multi-speaker version learning embeddings on a fixed set of speaker IDs. This can prove useful to ease potential speaker-specific integration.

2.4 About Text-to-Speech

Before developing a text-to-speech model, we first have to identify what can and cannot be done by TTS methods. The main challenge of TTS generation relies on the fact that it is a one-to-many mapping task. Indeed, the text context is a poorer description of speech, and many other *modalities* impact the output speech. This unsolvable task cannot be fully addressed as there is no single "good answer". However, we can hope to add some controllability over the output of our model to get closer to how an ideal TTS would work (see Fig. 4).

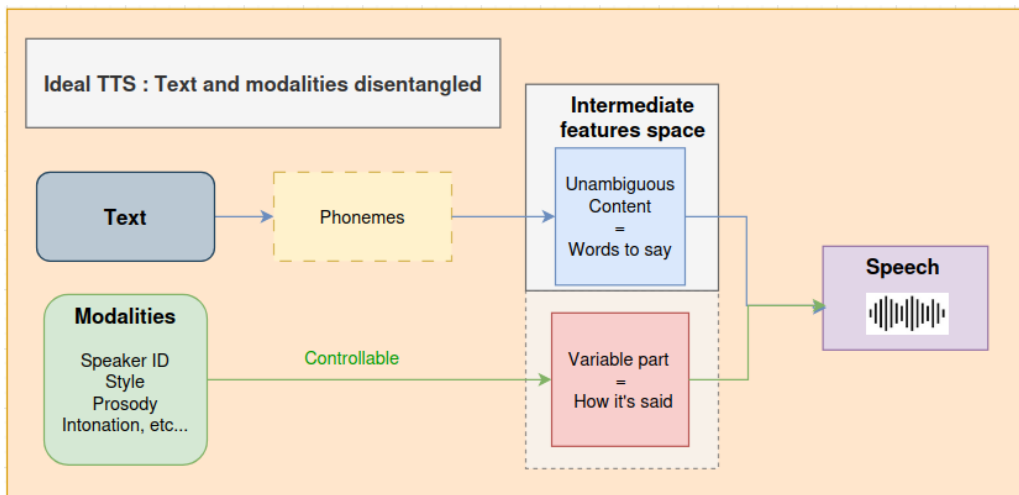


Figure 4: Ideal TTS

Regarding the ideal TTS objective, some modalities seem easier to manage than others. Focusing on the speech production process (see Fig. 5), we can distinguish modalities that are more contextual and others that are more individual. In the current TTS paradigm, the generation step mostly happens at the acoustic level. As we can see, this level is at the very end of the speech production stream, making it difficult to condition synthesis on the previously mentioned modalities. However, the articulatory trajectories, as defined by SPARC, are located on an upper level. While they fall short in controlling all the modalities, they do open the possibility to condition synthesis on more individual variability factors. It is important to understand that Fig. 5 is a simplification and that the mentioned modalities are in reality not totally separated. Moreover, one of the main weaknesses of the articulatory space is its reliance on source features that depend on more than the sheer phonological content.

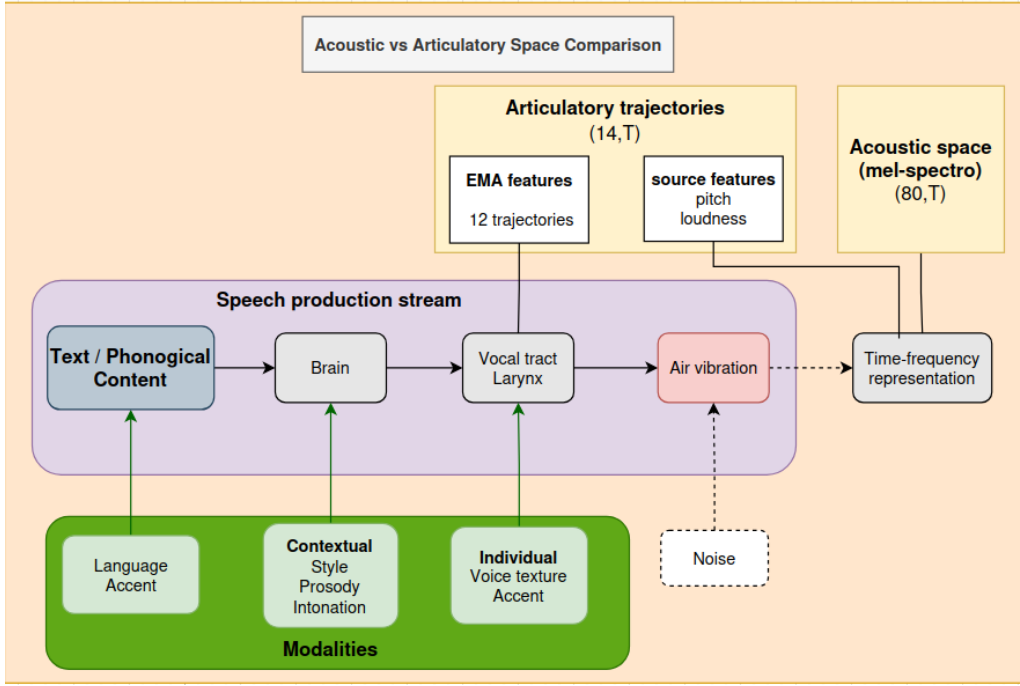


Figure 5: Articulatory vs Acoustic space comparison regarding the speech production process

2.5 Research Directions

The overall objective of this internship is to design, implement, and evaluate an **articulatory-based text-to-speech model** for controllable speech generation. Beyond model development, the project aims to investigate the universality of SPARC articulatory representations, their interest compared to acoustic representations, and their impact on speech quality across speakers and languages.

This research follows a progressive approach to articulatory-based TTS development, beginning with **SPARC articulatory space evaluation** to assess the SSL-linear AAI inversion method on datasets beyond MNGU0, quantifying performance degradation across different conditions, while accounting for inherent EMA data noise. This foundation enables the **development of monolingual articulatory TTS systems** that compare articulatory control spaces against mel-spectrogram-based synthesis, investigating whether proximity to ground-truth trajectories translates into improved speech quality, through correlation analysis between objective metrics and perceptual measures.

Building upon monolingual results, the research advances to **multilingual multispeaker TTS extension**, adapting the articulatory framework using speaker embeddings, while analyzing their robustness across speaker identities, recording conditions, and dataset variability. This progression from monolingual to multilingual synthesis allows for systematic evaluation of stability and generalization capabilities across languages and speakers.

3 SPARC multilingual AAI probing

The main objective of this section is to assess whether it is wise to reuse SPARC SSL-linear system to generate articulatory targets for our articulatory-based text-to-speech model. In particular, we want to evaluate the assumption that SPARC acoustic-to-articulatory inversion fitted on MNGU0 EMA dataset is indeed equivalent, up to a geometric linear transformation, to capturing articulatory

information of any other speaker from any other language.

This is a bit different from what has previously been done. Indeed, SPARC [7] only evaluates multilingual generation at the speech level and not at the articulatory stage. In another paper, Cho et al. did study articulatory transferability when tuning SSL-linear AAI for different SSL pretrained on various language corpora on different EMA datasets (various language-speaker configurations) [9]. An interesting finding being that Italian speakers have alleged higher transferability. However, it is easier to reuse MNGU0-based template to benefit from SPARC framework and its readily available articulatory HiFi-GAN. This is why we specifically want to probe the transferability of SPARC set up with WavLM 9th layer tuned on MNGU0. We are interested in quantifying the drop in PCC score, if there is one, to motivate our ambition to build a universal TTS model for multilingual generation.

3.1 Electromagnetic Articulography (EMA) datasets

There are indeed very few speech Electromagnetic Articulography (EMA) datasets available. SPARC authors evaluated on MOCHA-TIMIT [16] and HPRC [30] English datasets. We looked for other languages dataset and managed to retrieve MPSKA (Italian) [5] and PB2007 (French) [4] but could not find EMA-MAE (English accented) or DKU-JNU-EMA (Mandarin) datasets. Since we also had MNGU0 [21] and MOCHA-TIMIT datasets, we tried to replicate SPARC results. Here is a brief description of the 4 EMA datasets that will also be used in the next parts as a ground truth reference.

- **MSPKA** (Italian)

Speakers: 3 (one male: cnz; two females: lls, olm) **Duration:** approximately 2 hours in total

Utterances: cnz (666), lls (629), olm (501) citation speech (read sentences)

Audio: 22,050 Hz **EMA:** 400 Hz

Notes: Audio/EMA misalignments identified and corrected: olm after sample 410; lls from samples 451–465. The lls recording yields lower results and was discarded in [9].

- **PB2007** (French)

Speaker: 1 (spk1) **Duration:** 22 minutes 57 seconds **Utterances:** 1109 samples, across four types, with some phonetical repetitions: Vowels (V) (39 / 14 unique), Vowel–Consonant–Vowel (VCV) (565 / 266 unique), Monosyllabic words (M) (395 / 200 unique), Short sentences (S) (110 / 110 unique)

Audio: 16,000 Hz **EMA:** 100 Hz

Notes : Non realistic speech except for the sentences, short spoken time compared to silences.

- **MOCHA-TIMIT** (British English)

Speakers: 6 in our dataset version (3 female: ffe0, fsew0, faet0; 3 male: maps0, msak0, mmjn0). SPARC used 7. **Duration:** around 30 minutes per speaker (SPARC claims 27 minutes)

Utterances: 460 sentences designed to include the main connected speech processes in English

Audio: 16,000 Hz **EMA:** 500 Hz

- **MNGU0** (British English)

Speaker: 1 male speaker **Duration:** approximately 75 minutes **Utterances:** 1312 samples

from newspaper reading (continuous citation speech)

Audio: 16,000 Hz **EMA:** 200 Hz

Notes: this dataset is widely used as a canonical template due to its high-quality and careful recording

3.2 Methodology

Preprocessing For all datasets, we applied a preprocessing pipeline to make articulatory and acoustic features compatible. The original EMA data was downsampled to 50 Hz to match the sampling rate of SSL features, and each channel was z-scored within utterances. We also upsampled EMA to 100 Hz using cubic spline interpolation, in order to have better granularity, but it did not show significant difference.

On the audio side, speech waveforms were processed through the 9th layer of WavLM SSL and the SPARC linear layer to produce synthetic EMA data in the MNGU0-speaker EMA space. The input audio was normalized to a 16 kHz sampling rate, with zero mean and unit variance.

Articulatory Transfer Method Following [9], we trained linear models to predict speaker-specific reference EMA traces from SPARC MNGU0 representation. A linear regression with L1 penalty (Lasso) was used to enforce sparsity in the mapping, with $\alpha = 0.01$, implemented with the `sklearn.linear_model.Lasso` library. The use of an L1 penalty limits the transformation to mainly geometrical differences. This procedure was repeated individually for each speaker of each dataset.

Train/Dev/Test Splits For most datasets, we applied a random split of 60/20/20 into train, development, and test sets. The PB2007 dataset required a tailored strategy due to its heterogeneous content, which includes vowels, vowel–consonant–vowel sequences, monosyllables, and short sentences. In this case, all vowels (39 tokens) and all VCVs (565 tokens) were included in the training set, while the 110 sentences were reserved exclusively for testing. The 395 monosyllables were distributed across train, dev, and test according to a parameter α that controls the share of monosyllables added to the train+dev pool, where the selected monosyllables were split evenly between training and development.

We tested this strategy for multiple values of $\alpha \in 0.01, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0$ and also ran an experiment including sentences in the train in order to evaluate to what extent shorter isolated utterances are sufficient compared to longer sentences in learning the cross-speaker transfer linear layer from SPARC AAI.

3.3 Results and Discussion

The results obtained are summarized in Fig. 1. We can first observe that we obtained relatively similar scores compared to SPARC experiments on MNGU0 and MOCHA, though a bit lower. This can be explained by some problematic samples that might have been filtered out (SPARC reports 27 minutes per speaker on average while we have around 30 minutes) or a different EMA data preprocessing ("A low-pass filter is applied to the features to remove high-frequency noise using the Butterworth filter of order 5, where the frequency threshold is set at 10 Hz") while I forgot to

smoothen the EMA traces.

Dataset (language)	Nb of speakers	SPARC reported PCC	Test PCC
MNGU0 (en)	1	0.878	0.855
MOCHA-TIMIT (en)	6	0.739	0.729
MSPKA (it)	3	-	0.714 (0.760*)
PB2007 (fr)	1	-	0.678 (0.747**)

Table 1: SPARC MNGU0-based AAI system on various EMA datasets - (*) when removing one of the 3 speakers scoring particularly low - (**) when including the test sentences in the training.

While the PCC scores are around 0.85 for the MNGU0 template speaker, the PCC scores on other datasets mainly range in the 0.70s (SPARC reported a 0.761 score for the HPRC English dataset). The universal transferability hypothesis is not completely verified, and there seem to be some unrecoverable differences that cannot be captured with a simple geometrical linear transformation. However, it does not seem critical, given the good speech resynthesis on various datasets achieved by SPARC. Besides, we are particularly satisfied that the cross-speaker transferability from SPARC representations does not seem to rely on the language used. Indeed, MSPKA and PB2007 are able to reach similar scores (around 0.75) to their English counterparts. For MSPKA, we reached a 0.76 PCC score when removing the lls speaker, which is of apparently poor quality (Cho et al. acted similarly in [9]). For PB2007, the obtained score is comparable when we include sentences in the training set (it can be seen as some sort of cheating, but otherwise it is a bit too big of a disadvantage considering that it is not learning on natural speech samples).

The particular experiment conducted on PB2007 shows that the speaker-specific linear transformation actually requires learning on sufficiently long speech trajectories. Indeed, we can observe in Fig. 6 that learning from short isolated phone samples only (vowels or vowel/consonant/vowel) reaches only a 0.58 score. While adding monosyllabic words allows us to gain around 10 points, as previously mentioned, including sentences achieves around 0.75 correlation. We also display scores when cropping head and tail silences. The fact that training scores are lower highlights the weakness of PCC scoring on short samples with lower variations.

We will retain a PCC correlation compared to ground truth EMA of around 0.75 as a baseline for the following sections featuring TTS articulatory prediction.

4 Monospeaker articulatory-based TTS

As a first step towards building our TTS model, we tried to adapt Grad-TTS for articulatory generation with minimal modifications. We wanted to compare the use of articulatory space vs. mel-spectrogram space as well as phonetic feature input vs. learned embedding. We also implemented variants of the goal phonetic features-to-articulatory model and compared them.

4.1 Data

The choice made to evaluate the adaptability of Grad-TTS was to first try it on a minimal difficulty setup. For this reason, we naturally reused the LJSpeech-1.1 dataset [28], which appears to be the

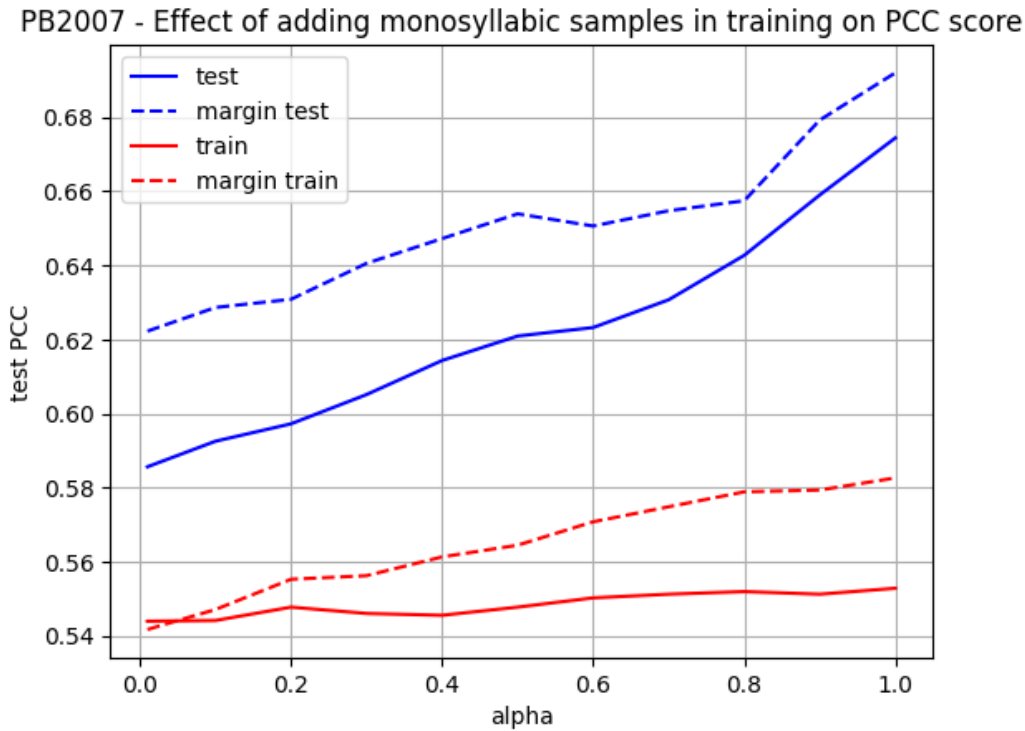


Figure 6: PCC scores improvement upon increasing the share of monosyllabic samples in the learning of the speaker-specific linear transformation. *Margin* refers to the cropping of head and tail silences

preferential one for TTS learning. It has the advantage of being a high-quality monolingual and monospeaker dataset. Unfortunately, it does not come with fully phonemized annotations or any form of phoneme alignment. To address the partial phonemization, we first improved the Grad-TTS text-to-phoneme converter to reach a higher rate of conversion (their version was maybe voluntarily imperfect, keeping a high rate of residual letters. For instance, a word next to a punctuation symbol would not be correctly parsed and retrieved in the text-to-phoneme dictionary). We achieved a full phoneme conversion for around 80

4.2 Methodology

Model Configurations

We want to compare multiple model configurations depending on the input type and the intermediate space but using the same generation backbone. For the intermediate space, we want to compare **articulatory-based vs. acoustic-based** versions. For the input space, we want to compare **phonetic ternary-features vs. vocabulary-based learned embedding**. For the latter, we distinguish **"text"** versions that reuse Grad-TTS imperfect text input processing, meaning that the text is partially converted to ARPabet phonemes with remnants of Latin-alphabet characters, and **"phnmtext"** versions that only process fully phonemized text input.

Given the fact that the train dataset is not aligned, we chose to keep the **duration predictor**. All the models developed use Grad-TTS as a feature generator, while the speech synthesis is done reusing either **SPARC HiFi-GAN English version** for articulatory models or Grad-TTS HiFi-GAN for acoustic models.

Here is a list of the different models.

- v1** : phonetic features / articulatory - The version we are interested in, (see 7)we also implemented a marginally different **v1_1** with loudness log-normalization.
- v2** : text or phnmtext / mel-spectro - Equivalent to Grad-TTS
- v3**: phonetic features / mel-spectro
- v4** : text or phnmtext / articulatory
- v5**: phonetic features / articulatory - Similar to **v1**, except that we tried to adapt the U-Net based score function with cross-channel attention with **v5** and **v5_preblock** versions.

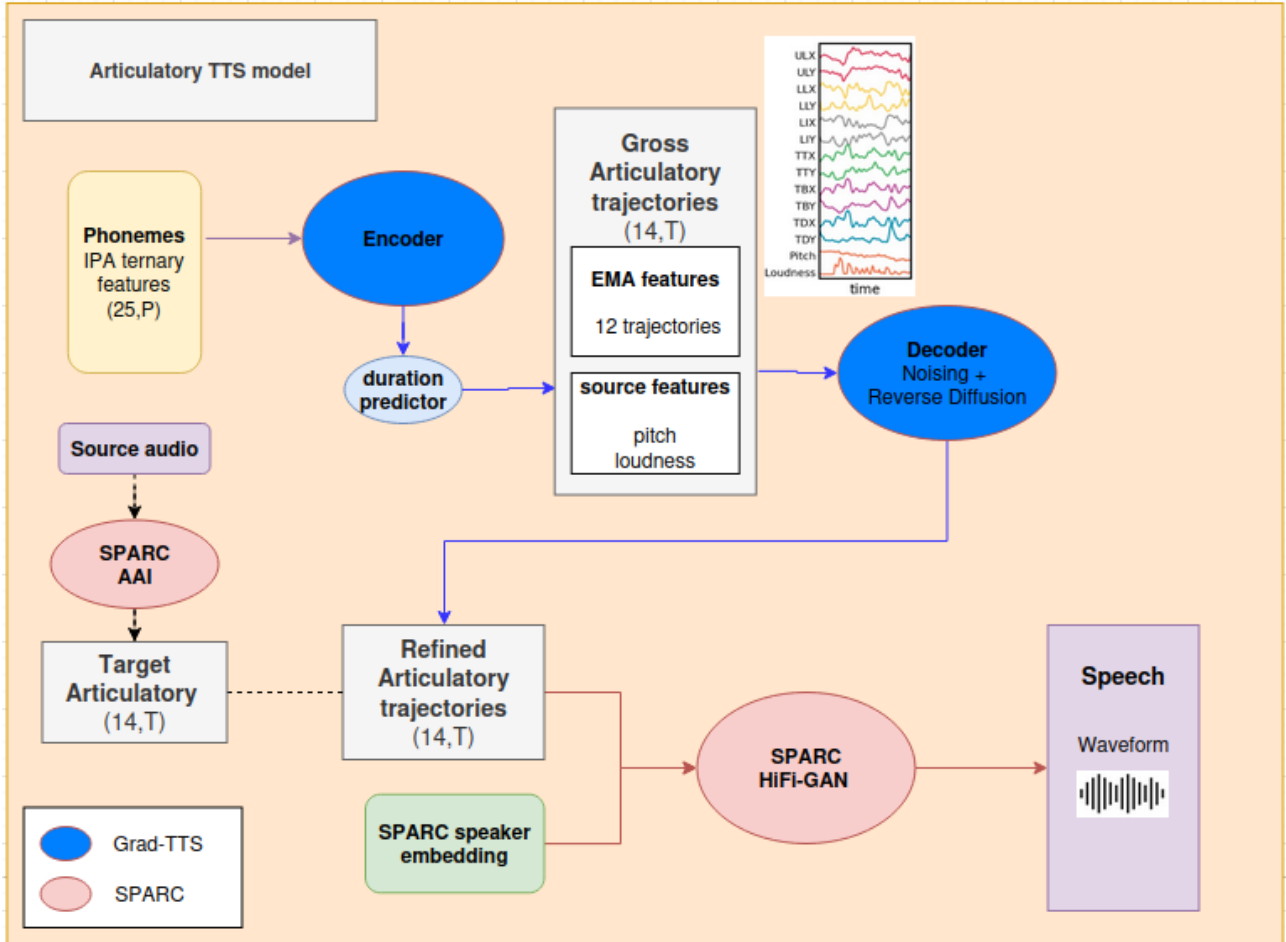


Figure 7: Developed TTS architecture

Implementation Details, Technical Specifications

The **phonetic features** are generated using Panphon conversion tables [2]. They provide, for every IPA phoneme, 24 phonological traits that are ternary-coded: 1 if present, 0 if unknown, -1 if absent. We use it to create 25-dimensional feature vectors, with the last dimension dedicated to silence encoding. It is much more compact and phonologically grounded, but less expressive and flexible than its **token-based** counterpart that learns a 192-dimensional embedding. Due to dimensionality compatibilities, the feature-based models only contain 1-head transformer blocks in their encoder, compared to 2 heads in Grad-TTS.

In the **intermediate space**, a few adaptations had to be done to accommodate articulatory trajectories. First, we had to expand the 14 articulatory channels to 16 to handle the downsampling scheme of the U-Net. We also chose to **reorder the channels** to improve spatial coherence on the

2D time-feature map. For instance, the tongue articulators are grouped together. The 2 additional channels are used to separate the rather independent pitch and loudness channels, with zero padding.

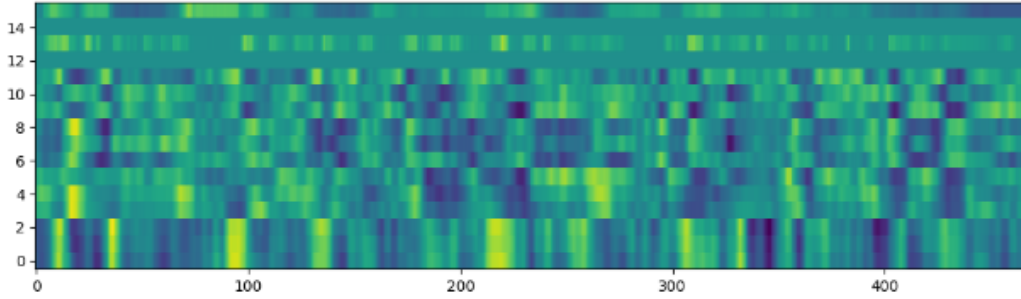


Figure 8: Articulatory channels reordering and padding

v5 versions While Grad-TTS provides a ready-to-work U-Net architecture for score function learning, it might not be totally adapted for our trajectory features. Indeed, it uses 2D convolutions as a way to generate features among all its constituent blocks. This forces features to be computed from neighbouring channels. Instead, we wanted the model to freely choose the channels that it deems useful for learning. For **v5**, we replaced all the (3,3) kernel Conv2D layers with (1,3) kernel Conv1D layers generating channel-wise features, combined with a cross-channel attention block. In order to guarantee a more stable version, we also implemented **v5_preblock**, where we did the same thing but only at a preliminary level to spatially rearrange input features for the unchanged Grad-TTS U-Net architecture.

All the models use the same hyperparameters as Grad-TTS: a 0.0001 learning rate with the Adam optimizer, a mini-batch size of 16, with the score function being learned on 2-second-long frames. It is important to note that only the "Grad-TTS" part (articulatory synthesis) is trained.

Comparison Methods While we implemented various versions to evaluate the contribution or limitations of input type or intermediate space choice, one of the main problems faced was the lack of comparability between our models. While all the speech outputs can be compared, it is difficult to compare quality between spectral and articulatory representations. Moreover, given the use of a duration predictor, we actually have output length issues, hindering us from even evaluating outputs from a single model compared to ground truth. The trick we found is to use Dynamic Time Warping (DTW). It is a widely used algorithm for measuring similarity between temporal sequences that may vary in length, but we are more interested in its capacity to align 2 sequences to matching lengths. However, DTW inherently distorts the temporal structure of the original data by stretching and compressing time segments to achieve optimal alignment. This can artificially inflate correlation metrics such as PCC scores, as the algorithm optimizes for similarity rather than preserving the original dynamics of the signals.

Hence, models within the same intermediate spaces will be quantitatively compared based on their DTW-based PCC results with ground truth. For articulatory generations, we will use the linear layers learned in the previous section to project to the corresponding speaker space. We will also compare the results obtained with aligned phoneme generation when available. The audio quality will be compared using UTMOS machine-based scoring [23] from this repository [24]. The generations will be evaluated on the previously mentioned datasets: MNGU0, MOCHA-TIMIT,

MSPKA, and PB2007. The speech quality will be evaluated on LJSpeech-1.1 as well. For text-based input models, we could not generate data for PB2007 as it does not provide text annotations. Lastly, the speech output generation actually requires a speaker embedding for the articulatory models. We unfortunately had to use the original audio with the SPARC speaker encoder, but we can imagine generating one using any speaker audio sample in case we do not have the source audio.

4.3 Results and Discussion

First, **v3** will not be included in the results, as its training did not converge, seemingly due to duration predictor alignment struggles. We attribute that to the fact that using ternary-phonetic features does not allow sufficient expressivity to generate complex mel-spectrograms. As a consequence, mel-spectrogram generations could not be compared (**v3** vs **v2**).

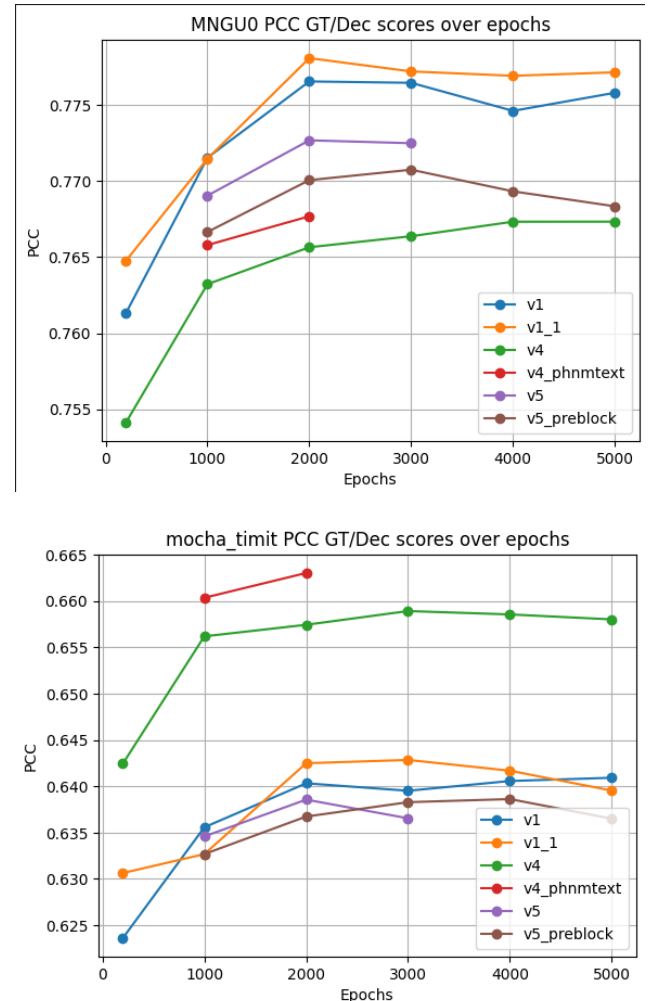
4.3.1 Articulatory Performance Analysis

Regarding the previous observation, only the articulatory models will be compared, meaning **v1**, **v4**, and **v5** versions. A first observation is that the scores lie in a relatively narrow range (around 0.01), especially when considering phonetic-features models (**v1** and **v5**). Generally speaking, the performances seem to be consistently better around 2000 epochs, which is comparable to Grad-TTS training (2300 epochs).

Overall, the **v1** and **v1_1** versions perform better, with a slight advantage for the latter log-normalized loudness version. The **v5** versions did not yield any notable improvement, probably because the articulatory channel reordering already groups mutually informative channels together on the time-articulatory 2D map. The **v4** versions are sometimes better, sometimes worse, but generally suffer from limited generalization to unknown vocabulary or phonemes.

Focusing on the **v4** versions, we observed consistent trends linked to phoneme vocabulary coverage. On MNGU0, the version exposed only to phoneme embeddings (*phnmtext*) outperformed the more letter-based version (**v4**), despite suffering from many out-of-vocabulary phonemes. This can be explained by a more refined learning of phoneme representations compared to hybrid letter/phoneme learning.

For MOCHA-TIMIT, where the phoneme vocabulary overlap with the training data is much higher (6 out-of-vocabulary phonemes compared



to 17 for MNGU0), v4 versions performed better than phoneme features versions. In this case, we suspect that using learned embeddings was particularly effective, as generalization on unseen phonemes was less critical.

In contrast, results on MSPKA were significantly worse for v4 versions compared to v1 versions, which was expected given the large number of out-of-vocabulary Italian phonemes. This observation supports the hypothesis that ternary phoneme features grant better generalization, especially to new languages. However, v4 still outperformed *phnmtext* here, as the use of letter embeddings allowed for partial cross-lingual transfer between Italian and English.

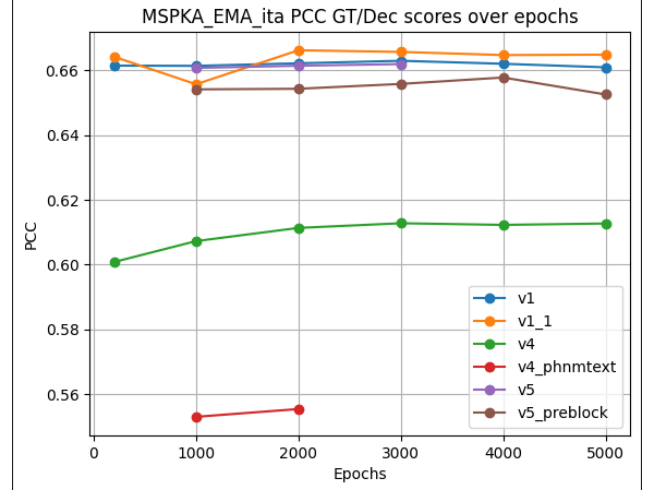


Fig. 2 focuses on the performance of **v1_1** after 2000 epochs. On average, it shows a drop of about 0.08 compared to the results reported in the previous section. This decrease is expected, as generating articulatory trajectories from text is inherently more difficult than from audio.

It is also important to note that the use of Dynamic Time Warping (DTW) for alignment artificially inflates the correlation (PCC) scores, meaning that the reported values may appear slightly better than the true performance. This is confirmed by the worse results obtained when running aligned phoneme inputs inference.

For comparison, we also evaluate against SPARC, the framework on which these models are optimized. The scores for **v1_1** are relatively stable around 0.75 across datasets, with the expected exception of the template MNGU0, which yields a higher score of 0.82. We also note that the correlations were computed on EMA traces only. The pitch and loudness required for speech reconstruction, which are actually harder to generate from phonemes, are left out of the correlation computation.

Dataset (language)	v1_1 /ground truth PCC	v1_1/sparc PCC
MNGU0 (en)	0.778	0.824
MOCHA-TIMIT (en)	0.642	0.757
MSPKA (it)	0.626 (0.666*)	0.745
PB2007 (fr)	0.597	0.754

Table 2: PCC scores summary of v1_1 generated EMA traces after 2000 epochs - (*) when removing lls speaker

4.3.2 Speech Synthesis Quality Analysis

Compared to the previous section, since all the models and datasets are now comparable at the speech level, we introduce the LJSpeech-1.1 dataset to evaluate generalization on the same dataset. We also include **v2** and drop the **v5** versions.

The results obtained after inference are summarized in Fig. 9. First, it is important to note that UTMOS scoring is not entirely reliable. Some irregularities are observed: for instance, the original audio from MOCHA and MSPKA receives relatively low scores despite being perceived as better when inspecting samples manually. Performance also appears generally worse on non-English datasets, which is consistent with the UTMOS paper [23], as the model was trained primarily on English data. We also suspect some overfitting on synthetic speech data, which may explain poor results on real audio. Nevertheless, UTMOS remains the only feasible method for large-scale, relatively neutral evaluation.

From the UTMOS scores, several trends can be noted. Despite resynthesizing from the original audio, SPARC synthesis does not consistently outperform text- or phoneme-input models. Among the phoneme-feature articulatory versions, **v1** appears slightly better than **v1_1**. For the learned embedding versions, **v4** performs slightly better overall, but **v2** is the most effective, suggesting that generating speech in the acoustic mel-spectrogram space may produce more realistic outputs. Overall, however, the results are not conclusive, and strong claims about relative model performance are difficult to make based on UTMOS alone.

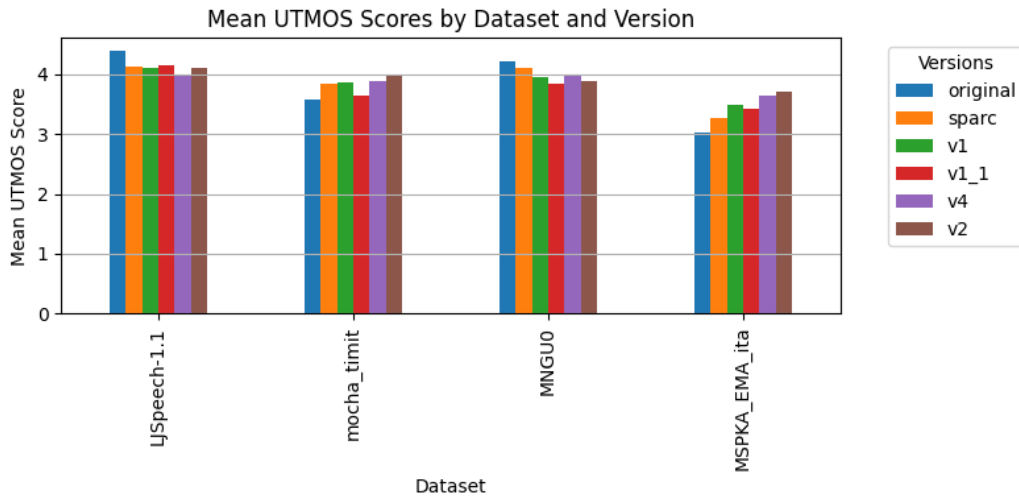


Figure 9: UTMOS model comparisons per dataset after 2000 epochs

UTMOS scores do not always align with perceptual evaluation from listening. For instance, Italian generations are largely unintelligible across all models and retain an English accent. This is expected, since the training data and the HiFi-GAN used for articulatory models were English. Otherwise, synthesized speech is generally intelligible, though quality varies in terms of prosody, intonation, and smoothness.

Listening confirms the UTMOS prediction differences between **v1** and **v1_1**: the log-normalized version (**v1_1**) feels less natural and less smooth. Prosody and intonation are particularly weak in phoneme-feature models, suggesting that learned embeddings better capture complex sequential patterns. It is also important to note that in articulatory models, pitch and loudness channels must be generated solely from phonemes. These factors explain why the **v2** version appears to produce the most natural-sounding speech.

Another observation (not shown in figures) is that, in a replication of Grad-TTS training, UTMOS scores steadily improve across epochs, up to 10,000, for all datasets. This suggests that longer

training can enhance perceived speech quality, though perceptible improvement during listening was not clear.

The main takeaway from this section is that, while **v1_1** slightly outperforms **v1** in articulatory trajectory correlation, **v1** is significantly better at the speech level. This is partly because the HiFi-GANs were not trained on the generated intermediate features of any TTS models; for articulatory models, the vocoder was tuned on SPARC audio features, with which our synthesized trajectories have a correlation around 0.75.

5 Multispeaker Multilingual Articulatory-based TTS

Building on the monolingual, monospeaker TTS model, which maintained adequate phoneme correlation (PCC) with ground truth SPARC encoding while producing intelligible synthesized speech, this section explores the challenges of extending the system to multilingual and multispeaker scenarios.

The monolingual model’s primary limitations were in prosody prediction and phoneme alignment, which are particularly challenging in cross-lingual synthesis. To address these issues, we introduce an enhanced approach leveraging a comprehensive speech dataset with aligned phoneme annotations. This methodological improvement addresses a key weakness of the previous model: the difficulty of reliably predicting prosodic patterns from phoneme sequences across different languages. Integrating aligned annotations provides more robust prosodic and temporal guidance, which is expected to significantly improve the system’s ability to perform multilingual and multispeaker synthesis effectively.

5.1 Data

The primary dataset for this multilingual multispeaker TTS system is VoxCommunis [1], which provides phoneme-aligned annotations for the larger CommonVoice dataset [3]. CommonVoice is a participatory platform where volunteers contribute speech recordings, resulting in a heterogeneous corpus with varying audio quality. Unlike controlled datasets such as LJSpeech, CommonVoice samples may contain background noise, different recording conditions, and inconsistent microphone quality due to its crowdsourced nature. While the platform implements a community-based validation system where users vote on sample quality, this filtering mechanism does not always ensure optimal audio standards. For this work, we utilized the "validated" subset, though some samples of questionable quality were still encountered.

Dataset Configuration We employed version *cv-corpus-16.1-2023-12-06* from the Jean Zay computing cluster, which includes 63 languages: 'Georgian', 'Japanese', 'Bashkir', 'Romanian', 'Hindi', 'Uzbek', 'Tatar', 'Greek', 'Serbian', 'Maltese', 'Yoruba', 'Belarusian', 'Ukrainian', 'Armenian', 'Slovak', 'Central Kurdish', 'Urdu', 'Turkish', 'Vietnamese', 'Albanian', 'Bulgarian', 'Tamil', 'Swedish', 'Basque', 'Indonesian', 'Swahili', 'Turkmen', 'Kurmanji Kurdish', 'Dhivehi', 'Hausa', 'Chinese (Hong Kong)', 'Bengali', 'Mongolian', 'Chinese (China)', 'Cantonese', 'Ligurian', 'French', 'Sorbian (Upper)', 'Chuvash', 'Korean', 'Dutch', 'Uyghur', 'Marathi', 'Abkhaz', 'Italian', 'Lithuanian', 'Slovenian', 'Kazakh', 'Punjabi',

'Russian', 'Czech', 'Guarani', 'Malayalam', 'Taiwanese (Minnan)', 'Thai', 'Portuguese', 'Kyrgyz', 'Polish', 'Catalan', 'Erzya', 'Hungarian', 'Kinyarwanda', 'Amharic'

Among these, we removed 4 languages due to insufficient sample quantities: Kazakh (kk), Amharic (am), Urdu (ur), and Albanian (sq).

We also kept 8 languages out of the training set to explore zero-shot generation. These were selected based on their linguistic isolation to evaluate the model's generalization capabilities: Basque (eu), Georgian (ka), Abkhazian (ab), Guarani (gn), Swahili (sw), Hausa (ha), Korean (ko), and Erzya (myv).

The training set was composed of the 51 remaining languages. We constructed our splits from the dataset's original splits that do not mix speakers.

Notably, English data was not available in the processed VoxCommunis release, likely due to processing delays associated with the high volume of English contributions to CommonVoice.

Articulatory Evaluation Data For articulatory prediction evaluation, we continued using the electromagnetic articulography (EMA) datasets from the previous monolingual experiments. These custom datasets (MSPKA, MNGU0, MOCHA, and PB2007) all feature aligned phoneme annotations, making them suitable for consistent evaluation across model iterations. Given the absence of English in the VoxCommunis training set, English effectively serves as an additional zero-shot evaluation language for our multilingual system.

5.2 Methodology

Model Architecture

The multilingual multispeaker model builds upon the previous monolingual architecture **v1_1** with several key modifications. The most significant change is the exclusive use of aligned inputs, **eliminating the need for the duration predictor component** from the original design. This simplification is enabled by the phoneme-aligned annotations provided in the VoxCommunis dataset.

Another modification we applied on the articulatory encoder is to generate conditionally on a speaker. **Speaker embedding integration** is motivated by recent findings from [9], which demonstrate that articulatory trajectories retain dialect-specific characteristics. To accommodate this, we incorporate speaker embeddings directly into the encoder for articulatory generation, hoping that it would capture dialectal articulatory patterns from the speaker's mother tongue. However, since our objective differs from SPARC's speaker embeddings, which were primarily designed for geometric considerations, we relearn **a dedicated feed-forward network (FFN)** on top of the speaker features to generate encoder-specific speaker embeddings.

This integration can be made smoothly from Grad-TTS's multispeaker architecture proposal, though our approach diverges from their learned speaker ID embeddings. We prioritize a dynamic speaker embedding system capable of generating representations for arbitrary speakers unseen during training, rather than fixed learned embeddings.

For vocoding, we transition from the English-only SPARC to the **multilingual SPARC HiFi-GAN vocoder combined with its speaker encoder**. The aligned input structure resolves output length prediction issues, theoretically allowing for custom HiFi-GAN training. While training a specialized

vocoder tuned to our model’s articulatory generations would have been preferable, time constraints necessitated using the existing SPARC framework.

We keep using **SPARC WavLM MNGU0 AAI targets** (monolingual SSL English speaker template space) for practical implementation reasons, leveraging the established SPARC framework and available vocoder. Though Italian speaker space and multilingual SSL-based AAI (XLS-R) might offer superior transferability [9], the existing infrastructure provides a robust baseline and a readily available vocoder.

Training Configuration

We conducted two distinct training experiments:

Monolingual multispeaker models: Two 20-hour datasets were trained separately, Italian (**v6**) and Chinese Mandarin (**v6_zhCN**). Italian was selected for its demonstrated high transferability [9] and the availability of ground truth EMA data (MSPKA). Chinese Mandarin was chosen for its linguistic distance from Italian and my understanding of it.

Multilingual model: Approximately 50 languages with a maximum of 1 hour per language, providing a comprehensive multilingual training foundation. We name it **msml1h** (meaning multispeaker multilingual 1h per language)

All models employ **identical hyperparameters** to Grad-TTS: Adam optimizer with a learning rate of 0.0001, mini-batch size of 16, and score function learning on 2-second frames. As in the previous section, only the Grad-TTS component (including the speaker encoding layer) undergoes training.

Evaluation Methodology

We use the same methods as in the previous section (PCC and UTMOS) to evaluate our models at the articulatory and speech levels. A single difference is that we will only be able to have PCC compared to SPARC ersatz generation when evaluating on VoxCommunis test sets. We will also have a look at pitch and loudness correlations as they actually play a role in speech synthesis.

The experimental design enables systematic evaluation across multiple dimensions:

Monolingual training establishes the performance ceiling achievable with this architecture and dataset combination.

Multispeaker analysis quantifies performance degradation compared to monolingual models while assessing zero-shot language generalization capabilities.

Cross-lingual transfer evaluation demonstrates the model’s ability to generalize beyond training languages.

5.3 Results and Discussion

The evaluation was conducted on VoxCommunis test sets (up to 1 hour per language) and four custom EMA datasets (MNGU0, MOCHA, MSPKA, PB2007). Monolingual models were trained for 5000 epochs, while multilingual models were trained for 4000 epochs, with the latter processing significantly more data per epoch due to the increased language diversity. It should be noted that UTMOS evaluations are unreliable for non-English datasets, limiting objective speech quality assessment for multilingual synthesis.

5.3.1 Articulatory Performance Analysis

First, for **monolingual models on VoxCommunis** test set, we can observe from Fig. 3 that compared to SPARC articulatory features, pitch correlation is significantly lower, while loudness generally reaches levels of correlation above 0.7. For every channel, the model scores better on the language it is trained on, with higher values for Italian than Chinese. The EMA correlations are good around 0.78 on the same language and maintain levels above 0.6 for foreign languages.

Test Language	v6 (it)			v6_zhCN		
	PCC EMA	PCC Pitch	PCC Loudness	PCC EMA	PCC Pitch	PCC Loudness
Italian	0.782	0.391	0.794	0.678	0.313	0.712
Chinese	0.596	0.194	0.694	0.773	0.351	0.779
French	0.652	0.335	0.738	0.621	0.295	0.696

Table 3: PCC scores compared to SPARC articulatory features for Monolingual Models at epoch 2000 on VoxCommunis dataset

For the **multilingual model on VoxCommunis** compared to SPARC, we first notice from Fig. 4 that a longer training does not provide substantial improvement, with the best scores around 1000–2000 epochs. The scores are lower than for monolingual models on the same language, especially for EMA (0.73 vs. 0.78). However, when looking at the zero-shot languages, the drop is less significant than for their monolingual counterparts. This supports the idea that the model benefits from the multilingual training and does learn universal representations.

Table 4: Multilingual Model Mean PCC values by Epoch: Training vs Zero-shot Languages

Epoch	PCC EMA		PCC Pitch		PCC Loudness	
	Training	Zero-shot	Training	Zero-shot	Training	Zero-shot
1000	0.737	0.686	0.385	0.335	0.771	0.721
2000	0.732	0.681	0.386	0.350	0.773	0.718
3000	0.728	0.672	0.392	0.355	0.780	0.727
4000	0.729	0.674	0.386	0.340	0.774	0.720

For the **custom datasets**, we can also compare to the ground truth EMA. The results (see Fig. 5) are overall surprisingly low on SPARC (< 0.5) compared to the scores we obtained for the VoxCommunis dataset. The results are even lower on ground truth. There is also some heterogeneity across the datasets.

5.3.2 Speech Synthesis Quality Results

We have the confirmation that the UTMOS scoring is not very significant for non-English data.

On the VoxCommunis data, the scores are indeed below 3 for every language, with an average score of around 2.5. For the monolingual models, each of them scores better on the language they were trained on.

Table 5: PCC results across custom datasets

Dataset	Ver sion	PCC generated over			
		gt_ema	sparc_ema	sparc_pitch	sparc_loudness
MNGU0 (en)	msml1h	0.388	0.443	0.545	0.556
	v6	0.356	0.411	0.480	0.539
	v6_zhCN	0.341	0.393	0.546	0.565
MSPKA (it)	msml1h	0.266	0.267	0.311	0.313
	v6	0.272	0.272	0.219	0.326
	v6_zhCN	0.231	0.240	0.352	0.341
MOCHA (en)	msml1h	0.267	0.335	0.303	0.546
	v6	0.251	0.314	0.373	0.539
	v6_zhCN	0.239	0.295	0.259	0.557

Dataset (language)	v6 (it)	v6_zhCN	msml1h
MNGU0 (en)	3.23	3.25	3.47
MOCHA-TIMIT (en)	3.31	3.05	3.37
MSPKA (it)	2.83	2.54	2.83

Table 6: UTMOS scores on UTMOS datasets at epoch 2000

On the custom dataset, we have the confirmation that the English language scores better on the UTMOS scale. Moreover, the scoring might still have some logic, as the Italian and multilingual models score better than the Chinese on MOCHA and MSPKA.

Regarding perceived speech quality assessed by audio listening, the first observation is that the sample quality seems very variable. Some samples appear intelligible, while others can hardly be listened to. Actually, even SPARC generations using the multilingual HiFi-GAN seem to produce chaotic samples, struggling with pitch. An interesting phenomenon is that monolingual models do have high residuals of mother-tongue accent, as if a foreigner tried to speak the language tested. The multilingual version tends to have a more or less neutral accent depending on the samples. The sound clarity could, however, be generally better. I have the feeling that lengthening the training tends to improve the smoothness of the audio, but not on all samples.

5.4 Discussion

Custom Dataset Performance Degradation. The custom EMA datasets exhibit consistently lower articulatory scores when compared to both SPARC predictions and ground truth trajectories. This degradation is particularly puzzling given that the model uses only phonetic inputs, which should theoretically maintain reasonable performance against SPARC comparisons regardless of audio quality differences.

The performance gap can be attributed to several factors. First, **SPARC trajectory quality mismatch** creates a fundamental discrepancy: while SPARC predictions on custom datasets are derived from high-quality audio recordings, our model’s articulatory predictions are learned from SPARC targets generated from lower-quality VoxCommunis audio. This quality differential introduces a

domain adaptation challenge that affects prediction accuracy.

Second, **speaker embedding domain mismatch** contributes to performance degradation. The speaker embeddings extracted from high-quality custom dataset audio may be out-of-distribution relative to the embeddings the model learned during training on VoxCommunis data. This mismatch could explain the reduced correlation with both SPARC and ground truth trajectories on custom datasets.

Importantly, the relatively good scores achieved on VoxCommunis may be artificially inflated due to overfitting on VoxCommunis-specific data. Since both training and evaluation use SPARC predictions derived from similar quality audio, the apparent success may mask underlying limitations that would become evident if ground truth articulatory data were available for VoxCommunis.

Monolingual versus Multilingual Comparison: Monolingual models consistently achieve superior articulatory scores compared to multilingual variants when evaluated against SPARC predictions. However, multilingual models demonstrate better generalization capabilities and maintain reasonable average performance across languages. The monolingual models also preserve **dialectal characteristics from the training language**, validating findings from [9] regarding the retention of language-specific articulatory patterns.

Articulatory-Speech Quality Decoupling: A critical finding is that **articulatory prediction accuracy (PCC) does not correlate strongly with speech synthesis quality**. This decoupling suggests that optimizing for trajectory accuracy may not directly translate to improved perceptual quality, indicating that other factors such as trajectory smoothness or realism may be more important for synthesis.

Vocoding Challenges: Even baseline SPARC speech synthesis exhibits quality limitations, with generally better UTMOS scores observed for English compared to other languages. The multilingual SPARC HiFi-GAN version demonstrates reduced stability compared to its English version. Interestingly, **proximity to SPARC predictions does not guarantee high-quality HiFi-GAN synthesis**, suggesting that the vocoder’s learned distribution may not align perfectly with our model’s output characteristics.

The TTS pipeline involves two distinct speaker embedding systems: the text-to-articulatory model uses embeddings learned from VoxCommunis data, while HiFi-GAN employs embeddings trained on SPARC data. This **speaker embedding mismatch** between synthesis stages may contribute to quality degradation, particularly when combined with suboptimal articulatory predictions.

Training Duration and Quality Relationship: Extended training (5000 epochs for monolingual models) appears to improve perceptual quality despite similar EMA correlation scores, suggesting that **trajectory smoothness and realism matter more than raw accuracy**. This indicates that HiFi-GAN may be better adapted to process smooth, realistic articulatory signals rather than trajectories that merely achieve high PCC scores.

Prosodic Limitations: The model demonstrates particular weakness in predicting pitch-related features while showing better performance for loudness prediction, which aligns with the expectation that loudness correlates more directly with phonemic content than pitch patterns. Multilingual models appear less capable of providing authentic language-specific prosodic cues, which is ex-

pected given the increased complexity of modeling diverse prosodic systems simultaneously.

Speaker Variability: Notable disparities exist between different speakers in synthesis quality, potentially attributable to speaker embedding quality variations or training data distribution imbalances across speakers within the multilingual framework.

5.5 Further discussion

Despite encouraging yet mixed results, the previous work raises new questions and opens up new research directions.

The multilingual multispeaker articulatory-based TTS system presents several avenues for investigation and improvement that merit further exploration. Understanding the contribution of different languages during multilingual training represents a critical research direction, as the current approach treats all languages equally without considering their potential differential contributions to cross-lingual generalization. A systematic study examining **how individual languages affect overall model** performance could inform more strategic language selection and weighting strategies for training data composition.

The potential for **fine-tuning the multilingual model on specific languages** is yet to be tested to evaluate the general model’s ability to fit on a language.

Speech quality enhancement emerges as a primary concern, particularly given the heterogeneous nature of the VoxCommunis dataset. Two complementary approaches warrant investigation: first, training a **custom HiFi-GAN vocoder** specifically optimized for the VoxCommunis articulatory and speaker embedding spaces could significantly improve synthesis quality by better matching the vocoder to the model’s intermediate distribution. Second, **combining articulatory and acoustic spectral features** in the intermediate representation space offers a promising direction, where articulatory features could serve as a regularization mechanism to guide acoustic feature learning and improve overall synthesis coherence. Another room for improvement might rely on better preprocessing of the training data by filtering out bad samples and audio signal denoising.

A fundamental question that requires deeper investigation concerns **the relationship between articulatory prediction accuracy and speech synthesis quality**. The correlation between accurate articulatory predictions, whether measured against ground truth EMA data or SPARC predictions, and the quality of resynthesized speech remains unclear. Understanding this relationship is crucial for determining whether improving articulatory accuracy necessarily translates to better perceptual quality or whether these two aspects can be optimized independently. The surprisingly good samples obtained from poor articulatory correlations also question the relevance of the PCC score.

The role of speaker embeddings in speech quality represents another critical area for investigation. While speaker embeddings are essential for multispeaker synthesis, their impact on overall speech quality and their interaction with articulatory features requires systematic analysis. Understanding how speaker representation affects both intelligibility and naturalness could inform better speaker modeling strategies. A first experiment could be to observe the results when combining speaker embeddings from good speech synthesis samples and synthesized good articulatory trajectories with poor speech quality from VoxCommunis.

The decision to **eliminate the duration predictor** in favor of aligned inputs, while simplifying

the architecture, raises questions about the potential benefits of explicit duration modeling for "real" TTS applications. Future work should examine whether reintroducing duration prediction capabilities could improve prosodic naturalness, particularly for languages with complex timing patterns that may not be fully captured by simple alignment. It might also improve pitch prediction, which seems to be a limiting factor for the current model.

Finally, a comparative analysis between **phonetic feature inputs versus learned embeddings** could provide insights into optimal input representation strategies. While phonetic features offer interpretability and linguistic grounding, learned embeddings might capture more nuanced linguistic patterns that improve synthesis quality. This comparison would help determine the most effective input representation for multilingual articulatory-based TTS systems.

6 Conclusion

This internship explored the development of articulatory-based TTS systems, progressing from monospeaker monolingual models to multilingual multispeaker architectures. We demonstrated that SPARC-derived articulatory representations can be effectively predicted from phonetic or text inputs, achieving reasonable correlation with ground truth EMA data and enabling intelligible speech resynthesis. Monolingual models generally outperform multilingual counterparts in articulatory accuracy, while multilingual models show better cross-lingual generalization.

Our experiments highlighted key challenges: pitch prediction remains difficult, speech quality does not strictly correlate with articulatory accuracy, and speaker embedding mismatches can degrade synthesis. The results also underscore the importance of aligned inputs, data quality, and vocoder adaptation.

Future work will focus on improving speech naturalness, exploring hybrid articulatory-acoustic representations, investigating speaker embedding strategies, and systematically analyzing cross-lingual transfer. Overall, this internship provides a solid foundation for articulatory-based TTS research and highlights avenues for advancing multilingual, multispeaker speech synthesis.

Acknowledgements

I would like to thank my internship supervisors at CoML lab, Emmanuel Dupoux and Angelo Ortiz Tandazo for closely supervising me and offering me the opportunity to work here. I would also like to thank all the team members for their warm welcoming, the PhD students Maxime Poli, Jing Liu, Jean-Baptiste Sevestre for the insightful discussions, the research engineers Tarek Kunze, Nicolas Hamilakis and Manel Khentout for their technical help and Lin Yang for helping me with the administration part. I would also like to thank my fellow interns Théo Charlot and Tom Fizycki for the good moments spent together. I would like to thank the MVA educational and administrative team for offering me the opportunity to follow high-level classes and do this internship. Finally, I would like to thank the Ecole des Mines de Paris professors Bruno Figliuzzi, David Ryckelynck and Elie Hachem for their teachings and their support that proved very helpful during this internship.

Glossary

AAI Acoustic-to-Articulatory Inversion. A modeling task that aims to predict articulatory movements (e.g., EMA trajectories) directly from speech audio. In this work, the AAI will often refer to the one developed by Cho et al. [8] [7] based on an SSL-linear AAI approach that constructs articulatory trajectories in the template canonical articulatory space of a single-speaker from the MNGU0 dataset. The motivation of projecting into a single articulatory space is based on the assumption that this template is universal in the sense that all the vocal tract anatomy geometries are equivalent up to an affine transformation and that the SSL-linear AAI purely captures phonological features and not voice identity information. . 4, 5

articulatory synthesis Articulatory synthesis aims to generate speech audio from articulatory features. I personally prefer the less ambiguous though longer articulatory-to-speech synthesis term.. 5

articulatory trajectories Time-varying kinematic patterns that describe the movements of speech articulators (such as the tongue, lips, and jaw) during speech production. In this work, they specifically refer to the x and y coordinates in the mouth mid-sagittal plane of 6 articulators : upper lip, lower lip, lower incisor, tongue tip, tongue blade and tongue dorsum. By extension, we will also include Pitch and Loudness signals under this term as they depict glottal excitation in the larynx and are necessary to reconstruct speech despite not being purely articulatory features. . 4

DTW Dynamic Time Warping. An algorithm for measuring similarity between temporal sequences that may vary in speed or timing. DTW finds an optimal alignment path between sequences by allowing flexible matching of time points, commonly used in speech processing and pattern recognition. While effective for sequence alignment, DTW can distort temporal structure and artificially improve correlation metrics. 17

EMA Electromagnetic Articulography. A technique for recording articulatory movements while speaking by tracking sensors attached to the tongue, lips, and jaw. In this work, we will sometimes refer to EMA data as the purely articulatory data (excluding pitch and loudness) i.e. the ones that can be acquired through EMA setup only.. 4–6

Grad-TTS A diffusion-based text-to-speech (TTS) model that generates speech from text or intermediate phoneme representations. Grad-TTS formulates speech synthesis as a denoising diffusion process, gradually transforming a noise sample into a realistic mel-spectrogram. The mel-spectrogram is then transformed to speech using a HiFi-GAN architecture [13] Here we are interested in adapting the model to the articulatory space.. 6

IPA International Phonetic Alphabet. A standardized system of phonetic notation used to represent speech sounds. In particular every phoneme can be characterized by list 24of phonological features based on there presence, absence or unknown status. These are referred as IPA-ternary features or traits.. 4

- MOS** Mean Opinion Score. A subjective measure of speech or audio quality, obtained by asking human listeners to rate the perceived naturalness or intelligibility of samples on a numerical scale (typically 1–5). In the context of speech synthesis, MOS is the standard evaluation metric for comparing the naturalness of generated speech to human speech or to competing models. Although widely used, MOS is costly to obtain since it requires human listening tests, and results may vary depending on experimental setup and listener population.. 6
- PCC** Pearson Correlation Coefficient. A statistical measure of linear correlation between two variables, used here to compare articulatory trajectories.. 6, 12
- SPARC** Speech Articulatory Coding. A neural framework for speech articulatory coding and articulatory-to-speech synthesis introduced by Cho et al. [7].. 4–6

References

- [1] Emily Ahn and Eleanor Chodroff. “VoxCommunis: A Corpus for Cross-linguistic Phonetic Analysis”. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. LREC 2022. Ed. by Nicoletta Calzolari et al. Marseille, France: European Language Resources Association, June 2022, pp. 5286–5294. URL: <https://aclanthology.org/2022.lrec-1.566/> (visited on 09/14/2025).
- [2] antoinelii. *Antoinelii/Panphon*. June 2, 2025. URL: <https://github.com/antoinelii/panphon> (visited on 09/11/2025).
- [3] Rosana Ardila et al. “Common Voice: A Massively-Multilingual Speech Corpus”. In: *Proceedings of the Twelfth Language Resources and Evaluation Conference*. LREC 2020. Ed. by Nicoletta Calzolari et al. Marseille, France: European Language Resources Association, May 2020, pp. 4218–4222. ISBN: 979-10-95546-34-4. URL: <https://aclanthology.org/2020.lrec-1.520/> (visited on 09/14/2025).
- [4] Pierre Badin et al. *PB2007 French acoustic-articulatory speech database*. Zenodo, Mar. 28, 2022. URL: <https://zenodo.org/records/6390598> (visited on 09/11/2025).
- [5] Claudia Canevari, Leonardo Badino, and Luciano Fadiga. “A New Italian Dataset of Parallel Acoustic and Articulatory Data”. In: *Interspeech 2015*. Interspeech 2015. ISCA, Sept. 6, 2015, pp. 2152–2156. DOI: 10.21437/Interspeech.2015-485. URL: https://www.isca-archive.org/interspeech_2015/canevari15_interspeech.html (visited on 09/11/2025).
- [6] Sanyuan Chen et al. “WavLM: Large-Scale Self-Supervised Pre-Training for Full Stack Speech Processing”. In: *IEEE Journal of Selected Topics in Signal Processing* 16.6 (Oct. 2022), pp. 1505–1518. ISSN: 1932-4553, 1941-0484. DOI: 10.1109/JSTSP.2022.3188113. arXiv: 2110.13900 [cs]. URL: <http://arxiv.org/abs/2110.13900> (visited on 04/15/2025).

- [7] Cheol Jun Cho et al. “Coding Speech through Vocal Tract Kinematics”. In: *IEEE Journal of Selected Topics in Signal Processing* 18.8 (Dec. 2024), pp. 1427–1440. ISSN: 1932-4553, 1941-0484. DOI: [10.1109/JSTSP.2024.3497655](https://doi.org/10.1109/JSTSP.2024.3497655). arXiv: [2406.12998](https://arxiv.org/abs/2406.12998) [eess]. URL: <http://arxiv.org/abs/2406.12998> (visited on 04/14/2025).
- [8] Cheol Jun Cho et al. “Evidence of Vocal Tract Articulation in Self-Supervised Learning of Speech”. In: *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. June 4, 2023, pp. 1–5. DOI: [10.1109/ICASSP49357.2023.10094711](https://doi.org/10.1109/ICASSP49357.2023.10094711). arXiv: [2210.11723](https://arxiv.org/abs/2210.11723) [eess]. URL: <http://arxiv.org/abs/2210.11723> (visited on 04/14/2025).
- [9] Cheol Jun Cho et al. *Self-Supervised Models of Speech Infer Universal Articulatory Kinematics*. Jan. 16, 2024. DOI: [10.48550/arXiv.2310.10788](https://doi.org/10.48550/arXiv.2310.10788). arXiv: [2310.10788](https://arxiv.org/abs/2310.10788) [eess]. URL: <http://arxiv.org/abs/2310.10788> (visited on 04/14/2025). Pre-published.
- [10] Jonathan Ho, Ajay Jain, and Pieter Abbeel. *Denoising Diffusion Probabilistic Models*. Dec. 16, 2020. DOI: [10.48550/arXiv.2006.11239](https://doi.org/10.48550/arXiv.2006.11239). arXiv: [2006.11239](https://arxiv.org/abs/2006.11239) [cs]. URL: <http://arxiv.org/abs/2006.11239> (visited on 09/10/2025). Pre-published.
- [11] Jaehyeon Kim et al. *Glow-TTS: A Generative Flow for Text-to-Speech via Monotonic Alignment Search*. Oct. 23, 2020. DOI: [10.48550/arXiv.2005.11129](https://doi.org/10.48550/arXiv.2005.11129). arXiv: [2005.11129](https://arxiv.org/abs/2005.11129) [eess]. URL: <http://arxiv.org/abs/2005.11129> (visited on 07/31/2025). Pre-published.
- [12] Jong Wook Kim et al. *CREPE: A Convolutional Representation for Pitch Estimation*. Feb. 17, 2018. DOI: [10.48550/arXiv.1802.06182](https://doi.org/10.48550/arXiv.1802.06182). arXiv: [1802.06182](https://arxiv.org/abs/1802.06182) [eess]. URL: <http://arxiv.org/abs/1802.06182> (visited on 09/11/2025). Pre-published.
- [13] Jungil Kong, Jaehyeon Kim, and Jaekyoung Bae. *HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis*. Oct. 23, 2020. DOI: [10.48550/arXiv.2010.05646](https://doi.org/10.48550/arXiv.2010.05646). arXiv: [2010.05646](https://arxiv.org/abs/2010.05646) [cs]. URL: <http://arxiv.org/abs/2010.05646> (visited on 04/15/2025). Pre-published.
- [14] Zhifeng Kong et al. *DiffWave: A Versatile Diffusion Model for Audio Synthesis*. Mar. 30, 2021. DOI: [10.48550/arXiv.2009.09761](https://doi.org/10.48550/arXiv.2009.09761). arXiv: [2009.09761](https://arxiv.org/abs/2009.09761) [eess]. URL: <http://arxiv.org/abs/2009.09761> (visited on 04/14/2025). Pre-published.
- [15] Sébastien Le Maguer, Simon King, and Naomi Harte. “The Limits of the Mean Opinion Score for Speech Synthesis Evaluation”. In: *Computer Speech & Language* 84 (Mar. 1, 2024), p. 101577. ISSN: 0885-2308. DOI: [10.1016/j.csl.2023.101577](https://doi.org/10.1016/j.csl.2023.101577). URL: <https://www.sciencedirect.com/science/article/pii/S0885230823000967> (visited on 09/08/2025).
- [16] *MOCHA-TIMIT*. URL: <https://www.cstr.ed.ac.uk/research/projects/artic/mocha.html> (visited on 09/11/2025).
- [17] Ethan Perez et al. *FiLM: Visual Reasoning with a General Conditioning Layer*. Dec. 18, 2017. DOI: [10.48550/arXiv.1709.07871](https://doi.org/10.48550/arXiv.1709.07871). arXiv: [1709.07871](https://arxiv.org/abs/1709.07871) [cs]. URL: <http://arxiv.org/abs/1709.07871> (visited on 04/15/2025). Pre-published.

- [18] Vadim Popov et al. *Grad-TTS: A Diffusion Probabilistic Model for Text-to-Speech*. Aug. 5, 2021. DOI: [10.48550/arXiv.2105.06337](https://doi.org/10.48550/arXiv.2105.06337). arXiv: [2105.06337 \[cs\]](https://arxiv.org/abs/2105.06337). URL: <http://arxiv.org/abs/2105.06337> (visited on 04/14/2025). Pre-published.
- [19] Ryan Prenger, Rafael Valle, and Bryan Catanzaro. *WaveGlow: A Flow-based Generative Network for Speech Synthesis*. Oct. 31, 2018. DOI: [10.48550/arXiv.1811.00002](https://doi.org/10.48550/arXiv.1811.00002). arXiv: [1811.00002 \[cs\]](https://arxiv.org/abs/1811.00002). URL: <http://arxiv.org/abs/1811.00002> (visited on 09/08/2025). Pre-published.
- [20] Yi Ren et al. *FastSpeech: Fast, Robust and Controllable Text to Speech*. Nov. 20, 2019. DOI: [10.48550/arXiv.1905.09263](https://doi.org/10.48550/arXiv.1905.09263). arXiv: [1905.09263 \[cs\]](https://arxiv.org/abs/1905.09263). URL: <http://arxiv.org/abs/1905.09263> (visited on 09/08/2025). Pre-published.
- [21] Korin Richmond, Phil Hoole, and Simon King. “Announcing the Electromagnetic Articulation (Day 1) Subset of the Mngu0 Articulatory Corpus”. In: Proc. Interspeech 2011. 2011, pp. 1505–1508. DOI: [10.21437/Interspeech.2011-316](https://doi.org/10.21437/Interspeech.2011-316). URL: https://www.isca-archive.org/interspeech_2011/richmond11_interspeech.html# (visited on 09/08/2025).
- [22] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. *U-Net: Convolutional Networks for Biomedical Image Segmentation*. May 18, 2015. DOI: [10.48550/arXiv.1505.04597](https://doi.org/10.48550/arXiv.1505.04597). arXiv: [1505.04597 \[cs\]](https://arxiv.org/abs/1505.04597). URL: <http://arxiv.org/abs/1505.04597> (visited on 12/13/2024). Pre-published.
- [23] Takaaki Saeki et al. *UTMOS: UTokyo-SaruLab System for VoiceMOS Challenge 2022*. June 29, 2022. DOI: [10.48550/arXiv.2204.02152](https://doi.org/10.48550/arXiv.2204.02152). arXiv: [2204.02152 \[cs\]](https://arxiv.org/abs/2204.02152). URL: <http://arxiv.org/abs/2204.02152> (visited on 08/19/2025). Pre-published.
- [24] *Sarulab-Speech/UTMOS-demo at Main*. Dec. 19, 2023. URL: <https://huggingface.co/spaces/sarulab-speech/UTMOS-demo/tree/main> (visited on 09/12/2025).
- [25] Jonathan Shen et al. *Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions*. Feb. 16, 2018. DOI: [10.48550/arXiv.1712.05884](https://doi.org/10.48550/arXiv.1712.05884). arXiv: [1712.05884 \[cs\]](https://arxiv.org/abs/1712.05884). URL: <http://arxiv.org/abs/1712.05884> (visited on 09/08/2025). Pre-published.
- [26] Jascha Sohl-Dickstein et al. *Deep Unsupervised Learning Using Nonequilibrium Thermodynamics*. Nov. 18, 2015. DOI: [10.48550/arXiv.1503.03585](https://doi.org/10.48550/arXiv.1503.03585). arXiv: [1503.03585 \[cs\]](https://arxiv.org/abs/1503.03585). URL: <http://arxiv.org/abs/1503.03585> (visited on 09/08/2025). Pre-published.
- [27] Yang Song et al. *Score-Based Generative Modeling through Stochastic Differential Equations*. Feb. 10, 2021. DOI: [10.48550/arXiv.2011.13456](https://doi.org/10.48550/arXiv.2011.13456). arXiv: [2011.13456 \[cs\]](https://arxiv.org/abs/2011.13456). URL: <http://arxiv.org/abs/2011.13456> (visited on 09/08/2025). Pre-published.
- [28] *The LJ Speech Dataset*. URL: <https://keithito.com/LJ-Speech-Dataset> (visited on 09/11/2025).
- [29] Peter Wu et al. *Deep Speech Synthesis from Articulatory Representations*. Sept. 13, 2022. DOI: [10.48550/arXiv.2209.06337](https://doi.org/10.48550/arXiv.2209.06337). arXiv: [2209.06337 \[eess\]](https://arxiv.org/abs/2209.06337). URL: <http://arxiv.org/abs/2209.06337> (visited on 07/28/2025). Pre-published.
- [30] Peter Wu et al. *Speaker-Independent Acoustic-to-Articulatory Speech Inversion*. July 24, 2023. DOI: [10.48550/arXiv.2302.06774](https://doi.org/10.48550/arXiv.2302.06774). arXiv: [2302.06774 \[eess\]](https://arxiv.org/abs/2302.06774). URL: <http://arxiv.org/abs/2302.06774> (visited on 09/11/2025). Pre-published.